



United International University

School of Science and Engineering

Department of Computer Science and Engineering

Mid Term Exam Trimester:- Fall 2025

Course Title: Fundamental Calculus

Course Code: MATH-1151 Marks:- **30** Time: **1** hour **30** minutes

[Note that the marks are given in brackets at the end of each question or part question. You have to answer all the questions in order.]

1. Given that, $f(x) = \frac{x+4}{x+3}$.
 - i) Find vertical and horizontal asymptotes. [2]
 - ii) Sketch $f(x)$. [2]
 - iii) Find the value of $\lim_{x \rightarrow -3} f(x)$. [1]
 - iv) Determine whether $f(x)$ is even, odd or neither. [1]
 - v) Find the domain and range of $f(x)$. [2]
2. Consider, $f(x) = 2(x - 1)^2 + 3$
 - i) The function has inverse for $x \geq a$ state the minimum value of a . [1]
 - ii) Find $f^{-1}(x)$. [2]
 - iii) On the same axis, plot $f(x)$ and $f^{-1}(x)$. [2]
 - iv) Given that, $g(x) = 3 - \sqrt{x - 3}$. Find $g \circ f$. [2]
3. (a) Consider the Following function:
$$f(x) = \begin{cases} 3x, & 0 \leq x < 1 \\ 3, & 1 \leq x < 2 \\ x^2, & x \geq 2 \end{cases}$$
 - i) Sketch the given function. [3]
 - ii) State whether the function is continuous at $x = 2$ or not. [1]
 - iii) State the domain and range of $f(x)$. [1]

P.T.O.

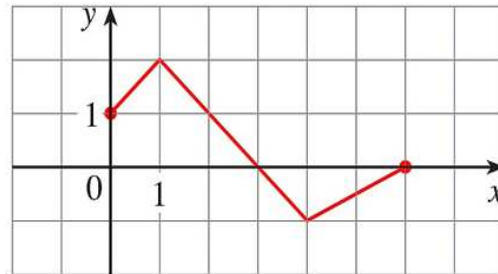
(b) Given that, $f(x) = 3 + 2e^{-x}$. Find the value of $\lim_{x \rightarrow \infty} f(x)$. [2]

4. (a) Given that, $f(x) = x^2 + 1$. Sketch $f(x)$.

Describe in words the transformation, $g(x) = 2f(x - 1) + 3$, and hence sketch it. [5]

(b) The graph of the function $f(x)$ is given as follows:

Figure 1:



i) The point $A(4, -1)$ is on the given graph. Find the reflection of point A about the x -axis. [1]

ii) Sketch $y = 2 - |f(x)|$. [2]